

Zero Order Mirror Descent Convergence

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In this excerpt we prove that Zero-Order Mirror Descent converges with high probability.

Assumption 1 f is ρ -Lipschitz continuous and convex over a closed convex set $Q \subset \mathbb{R}^d$.

Definition 1 $w : Q \rightarrow \mathbb{R}^d$ is a DGF if it is 1-strongly convex w.r.t. $\|\cdot\|$, in symbols, $w(y) \geq w(x) + \langle \nabla w(x), y - x \rangle + \frac{1}{2}\|y - x\|^2$. The Bregman divergence is defined

$$V_x(y) = w(y) - \langle \nabla w(y), y - x \rangle - w(x)$$

We have $V_x(x) = 0$ and $V_x(y) \geq \frac{1}{2}\|x - y\|^2 \geq 0$.

Classic examples are $w(x) = \frac{1}{2}\|x\|_2^2$ for the ℓ^2 norm which gives $V_x(y) = \frac{1}{2}\|y - x\|_2^2$, and $w(y) = \sum_i y_i \log(y_i)$ for the ℓ^1 norm with $Q = \Delta$, which gives KL divergence $V_x(y) = \sum_i y_i \log(y_i/x_i) \geq \frac{1}{2}\|x - y\|_1^2$.

1 Classic Mirror Descent

Definition 2 (Mirror Descent Step) The mirror descent step with step length α is

$$x^+ = \text{Mirr}_x(\alpha \nabla f(x)) \quad \text{where} \quad \text{Mirr}_x(\xi) = \underset{y \in Q}{\text{argmin}} \{V_x(y) + \langle \xi, y - x \rangle\}.$$

Mirror descents core lemma is this:

Lemma 3 If $x_{t+1} = \text{Mirr}_{x_t}(\alpha \nabla f(x_t))$, then

$$\forall u \in Q, \quad \alpha(f(x_t) - f(u)) \leq \alpha \langle \nabla f(x_t), x_t - u \rangle \leq \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + V_{x_t}(u) - V_{x_{t+1}}(u).$$

Proof Notice

$$\alpha \langle \nabla f(x_t), x_t - u \rangle = \langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + \langle \nabla f(x_t), x_{t+1} - u \rangle$$

Since Q is convex and $x_{t+1} = \text{Mirr}_{x_t}(\alpha \nabla f(x_t))$ is minimal, it satisfies the first order condition

$$\langle \nabla V_{x_t}(x_{t+1}) + \alpha \nabla f(x_t), u - x_{t+1} \rangle \geq 0 \quad \forall u \in Q$$

Therefore the above is at most

$$\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + \langle -\nabla V_{x_t}(x_{t+1}), x_{t+1} - u \rangle$$

Using the three-point identity

$$\langle -\nabla V_x(y), y - u \rangle = V_x(u) - V_y(u) - V_x(y)$$

we get the above equals

$$\begin{aligned} & \langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + V_{x_t}(u) - V_{x_{t+1}}(u) - V_{x_t}(x_{t+1}) \\ & \leq \left(\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle - \frac{1}{2} \|x_t - x_{t+1}\|^2 \right) + (V_{x_t}(u) - V_{x_{t+1}}(u)) \\ & \leq \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + (V_{x_t}(u) - V_{x_{t+1}}(u)) \end{aligned}$$

After recalling that $V_x(y) \geq \frac{1}{2} \|x - y\|^2$, using Cauchy-Schwarz $\langle a, b \rangle \leq \|a\| \cdot \|b\|$ and Young's inequality $2ab \leq a^2 + b^2$ in that order.

The first inequality follows because by convexity $f(u) \geq f(x_t) + \langle \nabla f(x_t), u - x_t \rangle$. $\blacksquare$

Using this, we can prove Mirror descent converges.

Lemma 4 For $\bar{x} = \frac{1}{T} \sum_{t=0}^{T-1} f(x_t)$, if we choose $\alpha = \sqrt{2\Theta/\rho^2 T}$, we have

$$f(\bar{x}) - f(x^*) \leq \rho \sqrt{\frac{2\Theta}{T}}$$

Proof Telescoping, we get

$$\sum_{t=0}^{T-1} \alpha (f(x_t) - f(x^*)) \leq \sum_{t=0}^{T-1} \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + V_{x_0}(x^*) - V_{x_T}(x^*)$$

Using convexity, $f(\bar{x}) \leq 1/T \sum_{t=0}^{T-1} f(x_t)$, and recalling that f is ρ -Lipschitz continuous, we have

$$f(\bar{x}) - f(x^*) \leq \frac{\alpha}{2} \rho^2 + \frac{1}{\alpha} \Theta,$$

where Θ is any number satisfying $\Theta \geq V_{x_0}(x^*)$. Now by choosing $\alpha = \sqrt{2\Theta/\rho^2 T}$ to balance the two terms, we are done. $\blacksquare$

2 Zero-Order Mirror Descent

Throughout the rest of this excerpt the norm $\|\cdot\|$ is the ℓ^2 norm and $Q = \mathbb{R}^d$.

Now, notice that the only place in the proof of Lemma 3 where we used that the update is evaluated at the true gradient $\nabla f(x_k)$ was to upper bound $f(x_k) - f(u)$. The gradient was not used in the proof, actually. So we get this lemma for free:

Lemma 5 If $x_{k+1} = \text{Mirr}_{x_k}(\alpha g_k)$, then

$$\forall u \in Q \quad \alpha \langle g_k, x_k - u \rangle \leq \frac{\alpha^2}{2} \|g_k\|^2 + V_{x_k}(u) - V_{x_{k+1}}(u).$$

Notice now that for any $u \in Q$,

$$\alpha(f(x_k) - f(u)) \leq \alpha\langle \nabla f(x_k), x_k - u \rangle = \alpha\langle \tilde{\nabla} f(x_k), x_k - u \rangle + \alpha\langle \nabla f(x_k) - \tilde{\nabla} f(x_k), x_k - u \rangle$$

Definition 6 (Zero-Order Estimator) *The Zero-Order gradient estimator is defined as follows*

$$\tilde{\nabla} f(x) = \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} \cdot p$$

where $p \sim \mathcal{N}(0, I_d)$.

For simplicity of exposition we will initially assume there is no quadratic error and $\tilde{\nabla} f(x_t) = p_t p_t^\top \nabla f(x_t)$. Thus the above equals,

$$\begin{aligned} & \alpha\langle \nabla \tilde{f}(x_t), x_t - u \rangle + \alpha\langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle \\ & \leq \frac{\alpha^2}{2} \|\tilde{\nabla} f(x_t)\|^2 + V_{x_t}(u) - V_{x_{t+1}}(u) + \alpha\langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle \end{aligned} \quad (1)$$

This lemma explains how to deal with the first term.

Lemma 7 *With probability at least $1 - \delta$, we have for every $1 \leq t \leq T$,*

$$|p_t p_t^\top \nabla f(x_t)| \leq c\sqrt{d} \log(cT/\delta)^{3/2} \|\nabla f(x_t)\|$$

Proof See Theorem 6 from EGGROLL convergence. ■

The main idea behind the rest of this proof is that, the mean-0 martingale error

$$\sum_{t=0}^{T-1} \alpha\langle (I - p_k p_k^\top) \nabla f(x_k), x_k - u \rangle$$

can hopefully be bounded by the other terms of the equation. In this case, it ends up being the “energy” term $V_{x_t}(u)$ and the Lipschitz constant ρ .

The main lemma which is different from the normal mirror descent bounds is the following.

Lemma 8 *Let $\{\mathcal{F}_t\}$ be a filtration. Assume that $g_t, d_t \in \mathcal{F}_{t-1}$, and assume $p_t \sim \mathcal{N}(0, I)$ with $p_t \in \mathcal{F}_t$, $p_t \perp \mathcal{F}_{t-1}$. Define*

$$E_\tau = \sum_{t=0}^{\tau-1} g_t^\top (I - p_t p_t^\top) d_t$$

Then for any θ , $0 < \delta < 1/e$, with probability at least $1 - \delta$,

$$|E_\tau| \leq c\theta \sum_{t=0}^{\tau-1} \log\left(\frac{c\tau}{\delta}\right) \|g_t\|^2 \|d_t\|^2 + \frac{1}{\theta} \log\left(\frac{c\tau}{\delta}\right).$$

Proof For each t let

$$A_t = \left\{ L_\delta \leq |p_t^\top d_t| \leq \sqrt{3 \log(c/\delta) \|d_t\|^2} \right\}, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(A_t) \geq 1 - \delta$$

Then by a lemma from before we know that $\mathbb{E}_{\mathcal{F}_{t-1}}[(I - p_t p_t^\top) d_t; A_t] = 0$. In particular,

$$\mathbb{E}_{\mathcal{F}_{t-1}} \left[g_t^\top (I - p_t p_t^\top) d_t; A_t \right] = 0.$$

Dropping the subscript t for simplicity,

$$\mathbb{E} \left[\exp \left(\frac{(g^\top p p^\top d \mathbb{1}_A)^2}{t^2} \right) \right] \leq \mathbb{E} \left[\exp \left(\frac{3 \log(C/\delta) \|d\|^2 (p^\top g)^2}{t^2} \right) \right]$$

Now, $p^\top g$ is $\mathcal{N}(0, \|g\|^2)$ random variable. Therefore, $\mathbb{E}[\exp((p^\top g)^2/4\|g\|^2)] \leq 2$. Now, taking $c \geq 1$ and $0 < \delta \leq 1/e$, we have

$$(g^\top (I - p p^\top) d)^2 \leq 2(g^\top d)^2 + 2(g^\top p p^\top d)^2$$

So, the random variable $Q_t = g_t^\top (I - p_t p_t^\top) d_t \mathbb{1}_{A_t}$ is $24\|g_t\|^2 \|d_t\|^2 \log(C/\delta)$ subgaussian and mean 0 given $\mathcal{F}_t$. Thus by the (subgaussian) Hoeffding inequality with probability at least $1 - \delta$ for any θ ,

$$\left| \sum_{t=0}^{\tau-1} Q_t \right| \leq c\theta \sum_{t=0}^{\tau-1} \log(c/\delta) \|g_t\|^2 \|d_t\|^2 + \frac{1}{\theta} \log\left(\frac{c}{\delta}\right)$$

Now $\sum_{t=0}^{\tau-1} Q_t = E_\tau$ on $\bigcap A_t$. Notice, since $\mathbb{P}(A_t) \geq 1 - \delta$, $\mathbb{P}(\bigcap A_t) \geq 1 - \tau\delta$. Therefore with probability at least $1 - (\tau + 1)\delta$, E_τ satisfies the same bound as above. Rescaling δ to $\delta/(\tau + 1)$ and we have completed the proof. $\blacksquare$

This above lemma is the key for making the argument work. Because $V_x(y) \geq \frac{1}{2}\|x - y\|^2$, and recalling that f is ρ -Lipschitz continuous, we have

$$\left| \sum_{t=0}^{\tau-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle \right| \leq \alpha \sqrt{c \sum_{t=0}^{\tau-1} \log^{3/2}(c\tau/\delta) \rho^2 V_{x_t}(u)}$$

after rescaling c and by choosing the optimal θ according to AM-GM $(x + y)/2 \geq \sqrt{xy}$. So, it remains to bound $\sum V_{x_t}(u)$. We can do this because Lemma 4 actually tells us a bound on $V_{x_\tau}(x^*)$.

So, by telescoping Equation (1), we arrive at the following lemma.

Lemma 9 Define $\chi = \log(cT^2/\delta)$. Then with probability at least $1 - \delta$, for every $1 \leq \tau \leq T$,

$$V_{x_\tau}(x^*) \leq \frac{c\tau d \chi^{3/2} \alpha^2}{2} \rho^2 + V_{x_0}(x^*) + \alpha \rho \sqrt{c \sum_{t=0}^{\tau-1} \chi^{3/2} V_{x_t}(x^*)}$$

Proof Telescoping Equation (1), we get

$$\begin{aligned} \sum_{t=0}^{\tau-1} \alpha(f(x_t) - f(x^*)) &\leq \sum_{t=0}^{\tau-1} \frac{\alpha^2}{2} \|\tilde{\nabla} f(x_t)\|^2 + V_{x_0}(x^*) - V_{x_\tau}(x^*) \\ &\quad + \sum_{t=0}^{\tau-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - x^* \rangle \end{aligned}$$

Therefore, by the previous discussion, by Lemma 7 and using that $f(x_t) \geq f(x^*)$ for any x_t ,

$$V_{x_\tau}(x^*) \leq \frac{c\tau d\chi^{3/2}\alpha^2}{2}\rho^2 + V_{x_0}(x^*) + \alpha\rho \sqrt{c \sum_{t=0}^{\tau-1} \log^{3/2}(c\tau/\delta) V_{x_t}(x^*)}$$

We can have this hold jointly for all $1 \leq \tau \leq T$ by rescaling $\delta \leftarrow \delta/T$. Then by using $\tau \leq T$, we arrive at the statement of the lemma. $\blacksquare$

Naturally, this above lemma should tell us how much $V_{x_t}(x^*)$ can grow. This next lemma tells us an upper bound on how much that is.

Lemma 10 *Let Θ be any upper bound on $V_{x_0}(x^*)$. Then for some absolute constant c ,*

$$\sup_{0 \leq t \leq T} V_{x_t}(x^*) \leq 2\Theta + cd\alpha^2\rho^2\chi^{3/2}T$$

Proof Define $M = \sup_{0 \leq t \leq T} V_{x_t}(x^*)$. Notice that the right hand side is increasing in τ . Therefore applying sup to both sides, we get

$$\begin{aligned} M &\leq \frac{cTd\chi^{3/2}\alpha^2}{2}\rho^2 + \Theta + \alpha\rho \sqrt{c \sum_{t=0}^{T-1} \chi^{3/2} V_{x_t}(x^*)} \\ &\leq \frac{cTd\chi^{3/2}\alpha^2}{2}\rho^2 + \Theta + \alpha\rho \sqrt{cT\chi^{3/2}M} \end{aligned}$$

Using Young's inequality, we know that

$$\alpha\rho \sqrt{cT\chi^{3/2}M} \leq \frac{1}{2}M + c\alpha^2\rho^2\chi^{3/2}T$$

Therefore,

$$\begin{aligned} M &\leq cTd\chi^{3/2}\alpha^2\rho^2 + 2\Theta + 2c\alpha^2\rho^2\chi^{3/2}T \\ &\leq 2\Theta + cd\alpha^2\rho^2\chi^{3/2}T \end{aligned}$$

for some absolute constant c as desired. $\blacksquare$

A natural question is: Is this a good bound at all? Shouldn't x_t be getting closer to the minimizer x^* ? But we showed $V_{x_t}(x^*)$ grows linearly?

To answer this, recall from before that $\alpha = \sqrt{2\Theta/\rho^2 T}$. Ignoring the polylog factor $\chi^{3/2}$, and dimension d , we have actually shown that $V_{x_t}(x^*)$ stays bounded. But still, shouldn't it

be decreasing to 0? Actually, it should not, since f could have many minimizers. The best bound you could hope to satisfy for every f is that it is just bounded.

For a similar choice of α , this tells us the accumulated error

$$\sum_{t=0}^{T-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - x^* \rangle$$

is around the same size as $V_{x_0}(x^*)$. This observation is what let's us prove convergence of Zero-Order mirror descent.

Theorem 11 (Zero-Order Mirror Descent Convergence) *Let $\bar{x} = 1/T \sum_{t=0}^{T-1} x_t$, if we choose*

$$\alpha = \sqrt{\frac{2\Theta}{\rho^2 c \chi^{3/2} T d}}$$

then with probability at least $1 - \delta$,

$$f(\bar{x}) - f(x^*) \leq \frac{1}{T\alpha} \cdot 9\Theta = \tilde{O}\left(\rho \sqrt{\frac{\Theta d}{T}}\right)$$

Proof By telescoping and using $V_{x_T}(x^*) \geq 0$, we get

$$\sum_{t=0}^{T-1} \alpha (f(x_t) - f^*) \leq \frac{cTd\chi^{3/2}\alpha^2}{2} \rho^2 + \Theta + \alpha \sqrt{c \sum_{t=0}^{T-1} \chi^{3/2} \rho^2 V_{x_t}(x^*)}$$

By using the previous lemma, we know that

$$\alpha \sqrt{c \sum_{t=0}^{T-1} \chi^{3/2} \rho^2 V_{x_t}(x^*)} \leq \alpha \sqrt{cT\chi^{3/2}\rho^2(2\Theta + c\alpha^2\rho^2\chi^{3/2}T)}$$

By choosing $\alpha = \sqrt{2\Theta/\rho^2 c \chi^{3/2} T d} \leq \sqrt{2\Theta/\rho^2 c \chi^{3/2} T}$ as $d \geq 1$, we know that

$$c\alpha^2\rho^2\chi^{3/2}T \leq 2\Theta$$

and therefore,

$$\alpha \sqrt{cT\chi^{3/2}\rho^2(2\Theta + c\alpha^2\rho^2\chi^{3/2}T)} \leq 4\Theta$$

Now again by our choice of α ,

$$\frac{cTd\chi^{3/2}\alpha^2}{2} \rho^2 \leq 4\Theta$$

Therefore,

$$\alpha(f(\bar{x}) - f^*) \leq \frac{9\Theta}{T}$$

We conclude,

$$f(\bar{x}) - f^* \leq \frac{9\Theta}{T\alpha} = \rho c \sqrt{\frac{9}{2}} \sqrt{\frac{d\Theta}{T}} \chi^{3/4} = \tilde{O} \left(\rho \sqrt{\frac{d\Theta}{T}} \right).$$

■

3 Relaxing The Quadratic Error Assumption

TODO